Candidate's Seat No:____________

Charotar University of Science & Technology(CHARUSAT)

First Semester of B. Tech. Theory Examination (ME/CL/EC/EE/IT/CE/CSE)

University Examination April-May, 2018

MA 141 Engineering Mathematics-I

Date:10/05/2018, Thursday **Time**: 01:30 p.m. to 04:30 p.m **Maximum Marks**: 70

INSTRUCTIONS:

1. **The question paper comprises two sections.**
2. **Section I and II must be attempted in separate answer sheets.**
3. **Make suitable assumptions and draw neat figures wherever required.**
4. **Use of scientific calculator is allowed.**

SECTION-I

Q.1 Select the correct option and write in the answer-book for each question given below. [07]

1. The n^{th} derivative of the function $\dfrac{1}{2x-3}$ is__________.

   (A) $\dfrac{(-1)^n n! 2^n}{(2x-3)^{n+1}}$

   (B) $\dfrac{(-1)^n (n-1)! 2^n}{(2x-3)^n}$

   (C) $\dfrac{(-1)^{n-1} (n-1)! 2^{n+1}}{(2x-3)^n}$

   (D) $\dfrac{(-1)^{n-1} (n-1)! 2^n}{(2x-3)^n}$

2. The n^{th} term in Maclaurin series expansion is __________.

   (A) $\dfrac{f^n(x)}{n!}$ (B) $\dfrac{f^n(0)}{n!}$ (C) $\dfrac{f(x)}{n!}$ (D) $\dfrac{f(0)}{n!}$

3. Which of the following function is differential over $(-1, 1)$.

   (A) $|x|$

   (B) $|x| + x^2$

   (C) $|x| - x^2$

   (D) x^2

4. If $u = \sin(x^2 + y^2)$ then $\dfrac{\partial u}{\partial x}$ is __________.

   (A) $2\cos(x^2 + y^2)$

   (B) $y\sin(x^2 + y^2)$

   (C) $2x\cos(x^2 + y^2)$

   (D) $\sin\left(\dfrac{y}{x^2 + y^2}\right)$

5. If $f(x, y) = x^2 + y^2$ is a function of two variable x and y, then total derivative of f is__________.

   (A) $2x + 2ydy$
   (B) $2xdx + 2ydy$
   (C) $2xdx + dy$
   (D) $2x + 2y$

6. For an implicit function $f(x, y) = c$, the value of $\dfrac{dy}{dx}$ is __________.

   (A) $\dfrac{f_x}{f_y}$
   (B) $\dfrac{f_y}{f_x}$
   (C) $-\dfrac{f_x}{f_y}$
   (D) $-\dfrac{f_y}{f_x}$

7. The equation of the tangent line to a surface $f(x, y, z) = 0$ at point x_0, y_0, z_0 is __________.

   (A) $(x - x_0)f_x(x_0, y_0, z_0) + (y - y_0)f_y(x_0, y_0, z_0) + (z - z_0)f_z(x_0, y_0, z_0)$
   (B) $(x - x_0)f_y(x_0, y_0, z_0) + (y - y_0)f_x(x_0, y_0, z_0) + (z - z_0)f_z(x_0, y_0, z_0)$
   (C) $(x - x_0)f_z(x_0, y_0, z_0) + (y - y_0)f_y(x_0, y_0, z_0) + (z - z_0)f_x(x_0, y_0, z_0)$
   (D) $(x - y_0)f_x(x_0, y_0, z_0) + (y - z_0)f_y(x_0, y_0, z_0) + (z - x_0)f_z(x_0, y_0, z_0)$

Q.2 Attempt any **four** of the following. [16]

(1) Obtain the n^{th} derivative of the function $\sin x \sin 2x \sin 3x$.

(2) Using Taylor's series, expand the polynomial $f(x) = x^5 + 2x^4 - x^2 + x + 1$ in powers of $x + 1$.

(3) If $y = (\sin^{-1} x)^2$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$.

(4) If $u(x, y) = \sin^{-1}\left(\dfrac{x + y}{\sqrt{x} + \sqrt{y}}\right)$, prove that

(i)$x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \dfrac{1}{2}\tan u$.

(ii)$x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x\partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = \dfrac{1}{4}(\tan^3 u - \tan u)$.

(5) If $u(x, y) = \log\left(\dfrac{x^3 + y^3}{xy}\right)$, verify that $\dfrac{\partial^2 u}{\partial x\partial y} = \dfrac{\partial^2 u}{\partial y\partial x}$.

(6) If $z = f(x, y)$ and $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$, show that $\dfrac{\partial z}{\partial u} - \dfrac{\partial z}{\partial v} = \text{x}\dfrac{\partial z}{\partial x} - y\dfrac{\partial z}{\partial y}$.

Q.3(a) Attempt any **two** of the following. [06]

(1) Find $\lim\limits_{x\to 0}\dfrac{2\sin x - \sin 2x}{x^3}$, if exists.

(2) Find the Jacobian $\dfrac{\partial(u, v)}{\partial(x, y)}$ for the functions, $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

(3) The period of a simple pendulum is $T = 2\pi\sqrt{\dfrac{l}{g}}$. Find the maximum error in T due to possible error up to 1% in l and 2.5% in g.

Q.3(b) Attempt any **one** of the following. [06]

(1) Discuss the maxima and minima of the function $x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$.

(2) Use Lagrange's method of undetermined multipliers to find the minimum values of $f(x, y, z) = x^2yz^3$, subject to the condition $2x + y + 3z = 1$.

SECTION-II

Q.4 Select the correct option and write in the answer-book for each question given below. [07]

1. The principal logarithm of any complex number $z = x + iy$ is __________.
   - (A) $\log \sqrt{x^2 + y^2} + \tan^{-1} \frac{y}{x}$
   - (B) $log\sqrt{x^2 - y^2} + i \tan^{-1} \frac{y}{x}$
   - (C) $log\sqrt{x^2 + y^2} - i \tan^{-1} \frac{y}{x}$
   - (D) $log\sqrt{x^2 + y^2} + i \tan^{-1} \frac{y}{x}$
2. If $z = x + iy$ is any complex number, then $\cosh ix =$ __________.
   - (A) $\sin ix$
   - (B) $\cos x$
   - (C) $\cos ix$
   - (D) $\sin x$
3. Which of the following statement is true?
   - (A) $Arg(-\sqrt{3} - i) = \frac{\pi}{3}$
   - (B) $Arg(-\sqrt{3} - i) = \frac{-5\pi}{6}$
   - (C) $Arg(-\sqrt{3} - i) = \frac{3\pi}{6}$
   - (D) $Arg(-\sqrt{3} - i) = \frac{\pi}{4}$.
4. If Σa_n and Σb_n is divergent then $\Sigma a_n + \Sigma b_n$ is __________.
   - (A) always divergent
   - (B) always oscillatory
   - (C) always convergent
   - (D) None of these
5. Which of the following series is not oscillate?
   - (A) $1 - 1 + 1 - 1 + ...$
   - (B) $1 - 2 + 3 - 4 + ...$
   - (C) $4 - 9 + 5 + 4 - 9 + 5 + 4 - 9 + 5 + ...$
   - (D) $-6 - 4 + 2 + 4 + ...$
6. The inverse of the matrix A is exists if __________.
   - (A) det(A)=0
   - (B) A is singular
   - (C) A is non-singular
   - (D) None of the above
7. If the homogeneous system has unique solution then the nullity of the homogeneous system is __________.
   - (A) 1
   - (B) 0
   - (C) 2
   - (D) None of the these

Q.5 Attempt any **four** of the following. [16]

(1) Solve the system of equations
$x - y + z = 1$
$2x + y - z = 2$
$5x - 2y + 2z = 5$
using Gauss elimination method.

(2) Determine for what values of λ and μ the following equations have
(i) no solution (ii) a unique solution (iii)infinite number of solutions
$x + y + z = 6$
$x + 2y + 3z = 10$
$x + 2y + \lambda z = \mu.$

(3) Reduce matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & -4 & 5 \end{bmatrix}$
in to row-echelon form and find its rank.

(4) Using Gauss-Jordan method, find the inverse of the matrix $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$, if exists.

(5) Find the root of the equations $x^5 = 1 + i$.

(6) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n^p}$.

Q.6(a) Attempt any **two** of the following. [06]

(1) Test the convergence of the series $\sum_{n=0}^{\infty} \frac{3^{3n}}{4^{2n}}$.

(2) Check whether the series $\frac{1}{2} + \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} + \ldots$ is convergent or divergent.

(3) Test the convergence of the series $1 - \frac{1}{2} + \frac{1}{3} - \ldots.$

Q.6(b) Attempt any **one** of the following. [06]

(1) Solve the equation
$x^3 - 6x^2 + 6x - 5 = 0$, using Cardan's method.

(2) Solve the equation $x^4 - 10x^2 - 20x - 16 = 0$ by Ferrari's method.

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